Quiz 1 Aug 25

Quiz 2 Oct 27

Prog. Group: Aug 17

Data File

Source

$X_1, X_2, \ldots X_n \overset{iid}{\sim}$ from $F \in \mathcal{F}$.

pth quantile: ξ_p

for a fixed $p \in (0,1)$, $H_0 : \xi_p = \xi_p^0$ vs $H_A : \xi_p \gneqq \xi_p^0$

$\hat{\xi}_p = X_{(r)}$ where $r = np \in I$

$[np]+1 \notin I$

$\psi_i = \psi(x_i - \xi_p^0) = 1$ if $x_i > \xi_p^0$

0 $\leq$

$\psi_1, \psi_2, \ldots \psi_n$ are iid Ber rv's with parameter

$$P(\psi_i = 1) = P(x_i > \xi_p^0)$$
$$= 1 - F(\xi_p^0)$$
$$= 1 - p \quad \text{under } H_0$$

Ranking Statistic:

$\text{rank}(x_i) = R_i$

$$T(R_1, R_2, \ldots R_n) = \sum_{i=1}^{n} a_i R_i$$

$$P(R_i = j) = \frac{1}{n}, \quad j = 1, 2, \ldots n$$

Signed-rank Statistic :

Exp: Tossing a coin twice

$\Omega = \{HH, HT, TH, TT\}$

a fair coin

$P(HH) = P(TT) = P(TH) = P(HT) = \frac{1}{4}$

X : # of heads in a trial

$X(\omega)$: 0 1 2

$$X^{-1}(-\infty, x] = \begin{cases} \phi & x < 0 \\ TT & 0 \le x < 1 \\ TT, HT, TH, & 1 \le x < 2 \\ \Omega, & x \geq 2 \end{cases}$$

$\Omega = \{a, b, c\}$ $\quad \mathcal{F} = \{\phi, a, \{b, c\}, \Omega\}$

$X(a) = 1, \; X(b) = 2, \; X(c) = 3$

↳ Not RV

$X_1, X_2, \ldots X_n$ iid obs. with cdf F_X, pmf p_X, pdf f_X

$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$

$X_{(1)} = \min\{X_1, X_2 \ldots X_n\}$

$X_{(r)} =$ rth smallest $\{X_1, X_2 \ldots X_n\}$, $r = 1(1)n$

$X_{(n)} = \max\{X_1, X_2, \ldots X_n\}$

$Y = X_{(1)}$

$$F_Y(y) = P(Y \le y) = P(X_{(1)} \le y) = 1 - P(X_{(1)} > y)$$

$$= 1 - \prod_{i=1}^{n} P(X_i > y)$$

$$= 1 - \prod_{i=1}^{n} \{1 - F_{X_i}(y)\}$$

$$= 1 - (1 - F_{X_i}(y))^n$$

$$p_Y(y) = F_Y(y) - F_Y(y^-)$$

$$f_Y(y) = \frac{d}{dy} F_Y(y) = n(1 - F_{X_i}(y))^n f_X(y)$$

$X_{(r)}$

$$F_{X_{(r)}}(x) = F_r(x) = P(\text{at least } r,\ X_i\text{'s} \le x)$$

$$= \sum_{j=r}^{n} P(\text{exactly } j,\ X_i\text{'s} \le x)$$

$$= \sum_{j=r}^{n} \binom{n}{j} (F_X(x))^j (1 - F_X(x))^{n-j}$$

$$= I_{F_X(x)}(r, n-r+1) = \frac{1}{B(r, n-r+1)} \int_0^{F_X(x)} u^{r-1} (1-u)^{n-r} \, du$$

$$\boxed{\begin{aligned} I(a,b) &= \frac{1}{B(a,b)} \int_0^1 u^{a-1} (1-u)^{b-1} \, du, \quad a, b > 0 \\ (0 < p < 1) \quad I_p(a,b) &= \frac{1}{B(a,b)} \int_0^p u^{a-1} (1-u)^{b-1} \, du \end{aligned}}$$

$$p_{X_{(r)}}(x) = F_{X_{(r)}}(x) - F_{X_{(r)}}(x^-)$$

$$= \frac{1}{B(r, n-r+1)} \int_{F(x-)}^{F(x)} u^{r-1} (1-u)^{n-r} \, du$$

$$f_{X_{(r)}}(x) = \frac{d}{dx} I_{F(x)}(r, n-r+1)$$

$$= \frac{1}{B(r, n-r+1)} F(x)^{r-1} (1-F(x))^{n-r} f_X(x)$$

Heuristic argument (Continuous case)

$$f_{X_{(r)}}(x) = \frac{d}{dx} F_{X_{(r)}}(x) = \lim_{h \to 0} \frac{F_{X_{(r)}}(x+h) - F_{X_{(r)}}(x)}{h}$$

$$= \lim_{h \to 0} \frac{1}{h} P(x < X_{(r)} \leq x+h)$$

$$= \frac{n!}{r!(n-r)!} (F_X(x))^{r-1} \lim_{h \to 0} \frac{F(x+h) - F(x)}{h} (1-F(x+h))^{n-r}$$

$$+ 0 + \dots$$

$$= \frac{1}{B(r, n-r+1)} (F_X(x))^{r-1} (1-F_X(x))^{n-r} f_X(x), \quad x \in \mathbb{R}$$

Date: 05-Aug-24

Joint Dist^n of $X_{(r)}, X_{(s)}$, $r < s$

$$F_{X_{(r)}, X_{(s)}}(x, y) = P(X_{(r)} \leq x, X_{(s)} \leq y)$$

$$= P(\text{at least } r \ X_k\text{'s} \leq x, \text{ at least } s \ X_k\text{'s} \leq y)$$

$$= \sum_{j=s}^{n} \sum_{i=r}^{j} P\Big(\text{exactly } \underset{(j-i)}{i} \ X_k\text{'s in } (-\infty, x], \ (x, y], \ \underset{n-j}{(y, \infty)}\Big)$$

$$= \frac{n!}{i!\,(j-i)!\,(n-j)!} (F(x))^{i} (F(y) - F(x))^{j-i} (1-F(y))^{n-j}, \quad x < y$$

$(x \geq y)$

$$P\left(X_{(n)} \leq x,\ X_{(s)} \leq y\right)$$

$$= P\left(X_{(s)} \leq y\right)$$

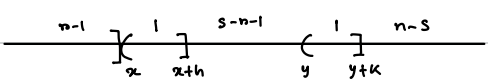
$$= \sum_{j=s}^{n} \binom{n}{j} \left(F(y)\right)^{j} \left(1 - F(y)\right)^{n-j}$$

pmf,

$$p_{X_{(n)}, X_{(s)}}(x, y) = F_{X_{(n)}, X_{(s)}}(x, y) - F_{X_{(n)}, X_{(s)}}(x, y-k) - F_{X_{(n)}, X_{(s)}}(x-h, y)$$

$$+ F_{X_{(n)}, X_{(s)}}(x-h, y-k)$$

$$f_{X_{(n)}, X_{(s)}}(x, y) = \frac{1}{hk} \lim_{\substack{h \to 0 \\ k \to 0}} P\left(x < X_{(n)} \leq x+h,\ y < X_{(s)} \leq y+k\right)$$



$$= \frac{n!}{(n-1)!\,(s-n-1)!\,(n-s)!} \left(F(x)\right)^{n-1} \lim_{h \to 0} \frac{1}{h}\left[F(x+h) - F(x)\right]$$

$$\times \left[F(y) - F(x+h)\right]^{s-n-1} \lim_{k \to 0} \frac{1}{k}\left[F(y+k) - F(y)\right]$$

$$\times \left[1 - F(y+k)\right]^{n-s}$$

$$= C_{n;s;n} \left(F(x)\right)^{n-1} \left(F(y) - F(x)\right)^{s-n-1} \left(1 - F(y)\right)^{n-s} f(x)\, f(y), \quad -\infty < x < y < \infty$$

Joint pdf of $X_{(r_1)}, \ldots, X_{(r_2)}$, $r_1 < r_2 < \cdots < r_k$

$$f_{X_{(r_1)} X_{(r_2)} \cdots X_{(r_k)}}(x_1, x_2 \ldots x_k) = \frac{n}{(r_1-1)(r_2-r_1-1)\cdots(r_k-r_{k-1}-1)(n-r_k)!} [F(x_1)]^{r_1-1} [F(x_2)-F(x_1)]^{r_2-r_1-1} \cdots [F(x_k)-F(x_{k-1})]^{r_k-r_{k-1}-1} [1-F(x_k)]^{n-r_k} f(x_1) f(x_2) \cdots f(x_k),$$

$$-\infty < x_1 < x_2 < \cdots < x_k < \infty$$

$$= n! \prod_{i=1}^{k} f(x_i) \prod_{i=1}^{k-1} \frac{[F(x_{i+1}) - F(x_i)]^{r_{i+1}-r_{i-1}}}{(r_{i+1}-r_{i-1})!} \frac{F(x_1)^{r_1-1} (1-F(x_k))^{n-r_k}}{(r_1-1)!\,(n-r_k)!}$$

Cont $X_1, X_2, \ldots X_n \overset{iid}{\sim}$ with cdf F, pdf f_X

<u>Joint of $X_{(1)}, X_{(2)} \ldots X_{(n)}$</u> : (Jacobian method)

$$X_{(1)} = g_1(X_1, X_2 \ldots X_n)$$
$$X_{(2)} = g_2(X_1, X_2, \ldots X_n)$$
$$\vdots$$
$$X_{(n)} = g_n(X_1, X_2, \ldots X_n)$$

$$f_{X_1, X_2 \ldots X_n}(x_1, x_2 \ldots x_n) = \prod_{i=1}^{n} f_X(x_i), \quad -\infty < x_i < \infty$$

$$f_{X_{(1)}, X_{(2)} \cdots X_{(n)}}(x_{(1)}, x_{(2)} \cdots x_{(n)}) = \sum_{\substack{\text{all cases of} \\ \text{inverse transformations}}} |J_i| \, f_{X_1, X_2 \cdots X_n}(x_{(1)}, x_{(2)}, \ldots x_{(n)})$$

$$= n! \prod_{i=1}^{n} f_X(x_{(i)})$$

$$f_{X_{(1)}, X_{(2)} \cdots X_{(n)}}(x_1, x_2 \ldots x_n) = n! \prod_{i=1}^{n} f_X(x_i)$$

Moments of order statistics :

1st order single moment :

$\forall r = 1(1)n, \quad \mu_{r:n} = E(X_{r:n})$

$$= \int x \, f_{r:n}(x) \, dx$$

$$= C_{r:n} \int x \, F(x)^{r-1} (1 - F(x))^{n-r} f(x) \, dx$$

$$|\mu_{r:n}| = \left| \int |x| \, f_{r:n}(x) \, dx \right| \leq \int |x| \, f_{r:n}(x) \, dx$$

(using Jensen's inequality)

$$\leq C_{r:n} \int |x| \, f(x) \, dx$$

Date: 06-Aug-25

If $E[|X|]$ exists then $E(X)$ finite.

$E(X_{r:n})$ is also finite.

If $\mu_{r:n}$ exists $\forall r = 1(1)n$

$$\sum_{r=1}^{n} \mu_{r:n} = \sum_{r=1}^{n} E(X_{r:n}) = E\left(\sum_{r=1}^{n} X_{r:n}\right)$$

$$= E\left(\sum_{r=1}^{n} X_r\right)$$

$$= \sum_{r=1}^{n} E(X_r) = n\mu$$

parallel converse :

$\mu_{r:n}$ finite for some $r = 1(1)n, n \geq 1$

but μ is not finite

Infinite moments must happen at extremes.

$$X_{1:n} < X_{2:n} < \cdots < X_{n:n}$$

$$\mu_{1:n} < \mu_{2:n} < \cdots < \mu_{n:n}$$

K th order sample moment

$$\mu_{r:n}^{(k)} = \int_{-\infty}^{\infty} x^k f_{r:n}(x)\, dx$$

$X \sim C(0,1)$

$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad x \in \mathbb{R}, \qquad F(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1} x, \quad x \in \mathbb{R}$

$$\mu_{r:n} = \frac{C_{r:n}}{\pi} \int x \left(\frac{1}{2} + \frac{1}{\pi} \tan^{-1} x\right)^{r-1} \left(\frac{1}{2} - \frac{1}{\pi} \tan^{-1} x\right)^{n-r} \frac{1}{1+x^2}\, dx$$

$\mu_{r:n}$ is finite $\quad r = 2(1)\, n-1, \quad n > 2$

$\mu_{r:n}^{(k)} \qquad r = k+1\,(1)\, n-k$

$$Var(X_{r:n}) = \sigma_{r:n}^2 = \sigma_{r,r:n}$$

$$= \mu_{r:n}^{(2)} - (\mu_{r:n})^2$$

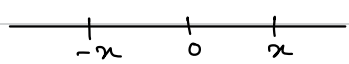
$(j,k)^{th}$ order product moment,

$$\mu_{r,s:n} = E(X_{r:n} X_{s:n}) = C_{r:s:n} \int_{-\infty}^{\infty} \int_{-\infty}^{y} xy \left(F(x)\right)^{r-1} \left(F(y) - F(x)\right)^{s-r-1} \left(1 - F(y)\right)^{n-r} f(y)\, f(x)\, dx\, dy$$

$$Cov(X_{r:n}, X_{s:n}) = \sigma_{r,s:n}$$

$$= \mu_{r,s:n} - \mu_{r:n}\, \mu_{s:n}$$

Dist^n symmetric about 0



$F(-x) = 1 - F(x)$

$X_1, X_2, \dots X_n \overset{iid}{\sim}$ cdf F, sym. about 0.

$$Y_i = -X_i, \quad \forall i = 1(1)n$$

$$Y \overset{d}{=} X$$

$$F_Y(x) = P(Y \le x) = P(-X \le x)$$

$$= P(X \geq -x)$$

$$= 1 - F_X(-x)$$

$$= F_X(x)$$

#

$X_{(1)} \dots X_{(n)}$

$Y_{(1)} \dots Y_{(n)}$

$$X_{(r)} \overset{d}{=} Y_{(r)} \overset{d}{=} -X_{(n-r+1)}$$

$$\left(X_{(1)}, X_{(2)} \dots X_{(n)}\right) \overset{d}{=} \left(-X_{(n)}, \dots -X_{(1)}\right)$$

$$\Rightarrow \mu_{r:n} = -\mu_{n-r+1:n}$$

$$\mu_{r:n}^{(k)} = (-1)^k \mu_{n-r+1:n}$$

$$\left(X_{(r)}, X_{(s)}\right) \overset{d}{=} \left(-X_{(n-s+1)}, -X_{(n-r+1)}\right)$$

$$\mu_{r,s:n} = (-1)^{1+1} \mu_{n-s+1,\, n-r+1:n}$$

$$\mu_{(r,s:n)}^{(j,k)} = (-1)^{j+k} \mu_{n-s+1,\, n-r+1:n}^{(j,k)}$$

$u = F(x), \quad x = F^{-1}(u)$

$= x(u)$

$$\mu_{r:n} = C_{r:n} \int_0^1 x(u)\, u^{r-1} (1-u)^{n-r}\, du$$

For $C(0,1)$, $\quad u = \frac{1}{2} + \frac{1}{\pi} \tan^{-1} x$

$$\tan^{-1}\left[\pi\left(u - \tfrac{1}{2}\right)\right] = x(u)$$

$$\Rightarrow \quad -\cot \pi u = x(u)$$

$$\mu_{r:n} = C_{r:n} \int_0^1 (-\cot \pi u)\, u^{r-1} (1-u)^{n-r}\, du$$

$$\boxed{\begin{array}{l} u = F(x) \\ v = F(y) \end{array}}$$

$$\mu_{r,s:n} = C_{r,s:n} \int_0^1 \int_0^v x(u)\, y(v)\, u^{r-1} (v-u)^{s-r-1} (1-v)^{n-s}\, du\, dv$$

$X_1, X_2 \ldots X_n, \quad U = F(X)$

$U_1 = F(X_1), \ldots, U_n = F(X_n)$

$X_{(1)} < X_{(2)} < \cdots < X_{(n)}$

$F(X_{(1)}) < F(X_{(2)}) < \cdots F(X_{(n)})$

$U_{(1)} < U_{(2)} < \cdots < U_{(n)}$

$U_{r:n}$

$$E[U_{r:n}] = C_{r:n} \int_0^1 u\, u^{r-1} (1-u)^{n-r}\, du = \frac{r}{n+1}$$

$$\int_0^\infty e^{-x} x^{p-1}\, dx = \Gamma p = (p-1)!$$

$$\int_0^\infty e^{-\alpha x} x^{p-1}\, dx = \frac{\Gamma p}{\alpha^p}$$

$$\# \quad \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1)$$

$$\frac{\sqrt{n}\,\left(g(\bar{X}_n) - g(\mu)\right)}{\sigma} \longrightarrow N\left(0, (g'(\mu))^2\right)$$

Date: 09-Aug-25

$X_1, X_2, \ldots X_n \overset{iid}{\sim} U(0,1)$

$X_{(1)} < X_{(2)} < \cdots < X_{(n)}$

$$f(x) = 1, \; 0 < x < 1$$
$$0, \; ow$$

$$u = F(x)$$
$$x = x(u) = F^{-1}(u) = u$$

$$\mu_{r;n}^{(k)} = C_{n;n} \int_0^1 u^k u^{n-1} (1-u)^{n-r}\, du$$

$$= \frac{B(r+k,\, n-r+1)}{B(r,\, n-r+1)} = \frac{\Gamma(r+k)}{\Gamma(n+k+1)} \frac{\Gamma(n+1)}{\Gamma(r)}$$

$k = 1, \; \mu_{r,n} = p_r$

$\sigma^2_{r;n} = \dfrac{p_r q_r}{n+2}$

$q_r = 1 - p_r$

$$\mu_{r,s;n} = C_{r,s;n} \int_0^1 \int_0^v u v \; u^{r-1} (v-u)^{s-r-1} (1-v)^{n-s}\, du\, dv$$

$$= \frac{r(s+1)}{(n+1)(n+2)}, \qquad \sigma_{r,s;n} = \frac{p_r q_s}{n+2}$$

#) Approximating moments of a.s. for any cont. Dist^n

$X_1, X_2, \ldots X_n \overset{iid}{\sim}$ from F

$F(X_1)\ F(X_2) \cdots F(X_n) \overset{iid}{\sim} U(0,1)$

$$U = F(X)$$

$X_{(1)} < X_{(2)} < \cdots < X_{(n)}$

$F(X_{(1)}) < F(X_{(2)}) < \cdots < F(X_{(n)})$

$U_{(1)} < U_{(2)} < \cdots < U_{(n)}$

$U_{(r)} = F(X_{(r)})$

$$X_{(r)} = F^{-1}(U_{(r)}) = g(U_{(r)})$$

$$\simeq g(P_r) + (U_{(r)} - P_r)\, g'(P_r) + \frac{(U_{(r)} - P_r)^2}{2!}\, g''(P_r)$$

$$E[X_{(r)}] \simeq g(P_r) \quad (1^{st} \text{ term})$$

$$= F^{-1}\left(\frac{r}{n+1}\right)$$

$$V[X_{(r)}] \simeq V[U_{(r)}]\left(g'(P_r)\right)^2 = \frac{p_r q_r}{n+2}\, \frac{1}{\{f(F^{-1}(P_r))\}^2}$$

$$= \frac{p_r q_r}{n+2}$$

$$g'(P_r) = \left.\frac{\partial g(U_{(r)})}{\partial U_{(r)}}\right|_{U_{(r)} = P_r}$$

$$= \left. \frac{1}{\frac{\partial U(r)}{\partial g(U(r))}} \right|_{U(r) = p_r}$$

$$= \left. \frac{1}{f(X(r))} \right|_{p_r}$$

$$= \left. \frac{1}{f[F^{-1}(U(r))]} \right|_{p_r}$$

$$= \frac{1}{f[F^{-1}(p_r)]}$$

$$Cov(X(r), X(s)) = \frac{p_r q_r}{n+2} \frac{1}{f\{F^{-1}(p_r)\}} \frac{1}{f\{F^{-1}(p_s)\}}$$

Asymptotic Distn of sample quantile

p th popln quantile ξ_p, $F(\xi_p) = p$

$X(r)$ is the p th sample quantile if $r = np \in I$
$[np]+1$

$$\frac{U(r) - p_r}{\sqrt{\frac{p_r q_r}{n+2}}} \xrightarrow{L} N(0,1) \qquad p_r = \frac{r}{n+1} \approx \frac{r}{n} = p$$

$$X(r) = g(U(r)) \qquad E(X_{r;n}) \simeq g(p_r) = F^{-1}\left(\frac{r}{n+1}\right)$$
$$= F^{-1}\left(\frac{r}{n}\right)$$
$$= F^{-1}(p) = \xi_p$$

$$\frac{X_{(n)} - \xi_p}{\sqrt{\frac{pq}{n}\frac{1}{\{f(\xi_p)\}^2}}} \longrightarrow N(0,1)$$

$$\frac{p_n q_n}{n+2} = \frac{pq}{n} \cdot \frac{1}{f(\xi_p)}$$

$X_1, X_2, \ldots X_n \overset{iid}{\sim} N(\mu, \sigma^2)$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$M_n \longrightarrow N\left(\mu, \frac{1}{4n}\sigma^2 2\pi\right)$$

$$\equiv N\left(\mu, \frac{\sigma^2\pi}{2n}\right)$$

Efficiency: $(M_n, \bar{X}_n)$ $= \dfrac{V(\bar{X}_n)}{V(M_n)} = \dfrac{\frac{\sigma^2}{n}}{\frac{\sigma^2\pi}{2n}} = \dfrac{2}{\pi}$

$\simeq 0.6366$

$X_1, X_2 \ldots X_n \overset{iid}{\sim} C(\mu, 1)$

$$M_n \xrightarrow{L} N\left(\mu, \frac{1}{4n}\pi^2\right)$$

$$f(x) = \frac{1}{\pi\left(1 + (x-\mu)^2\right)}$$

$$f(\mu) = \frac{1}{\pi}$$

Efficiency: $(M_n, \bar{X})$

$$= \frac{\infty}{\pi^2/4n} = \infty$$

Testing of hypothesis of $\mathcal{E}p$ fixed $p \in (0,1)$

$H_0: \mathcal{E}p = \mathcal{E}p^0$ (fixed)

vs $H_A: \mathcal{E}p \gtrless \mathcal{E}p^0$

Date: 11-Aug-25

Ex:

5. $X_1, \cdots\cdots, X_n$ is a random sample drawn from an exponential distribution with mean 10. What is the probability that the sample median is between 5 and 10 ?

chapter-2
Gibbons book pdf

$\rightarrow \quad P[a \le X_{(r)} \le b] = F_{X_{(r)}}(b) - F_{X_{(r)}}(a)$

$$= \sum_{j=r}^{n} \binom{n}{j} (F_X(b))^{n} (1-F_X(b))^{n-r} - \sum_{j=r}^{n} \binom{n}{j} (F_X(a))^{n} (1-F_X(a))^{n-r}$$

\# $U_1, U_2 \ldots U_n \overset{iid}{\sim} U(0,1)$

$R = U_{(n)} - U_{(1)} \sim B_I(n-1, 2)$

$QR = U_{(s)} - U_{(r)} \sim B_I(s-r, n-(s-r)+1)$

Ex:-

7. **(a)** Let X have a geometric distribution with pmf
$P(X = k) = (1-p)p^k;\ k = 0, 1, 2, \cdots\cdots;\ 0 < p < 1.$
Find the marginal distributions of $X_{1:n}$ and $X_{n:n}$ if $X_1, \cdots\cdots, X_n$ is a random sample from the above distribution.
(b) How would you modify part **(a)** if $X_1, \cdots\cdots, X_n$ have independent but non-identical geometric distribution with pmf
$P(X_i = k) = (1-p_i)p_i^k;\ k = 0, 1, 2, \cdots\cdots;\ 0 < p_i < 1;\ i = 1, 2, \cdots\cdots, n$?

$P(X_{(1)} = x) = F_{X_{(1)}}(x) - F_{X_{(1)}}(x-1)$

Ex:-

10. (a) Let $X_1, \cdots\cdots, X_n$ be a random sample from a population having absolutely continuous cdf $P(\cdot)$ and pdf $p(\cdot)$. Show that conditional distribution of $X_{(r)}$ given $X_{(s)} = x_s$, $r < s$, is same as the distribution of the rth order statistic in a random sample of size $(s-1)$ drawn from the population which is truncated to the right of x_s.

(b) Let $X_1, \cdots\cdots, X_n$ be a random sample from a population having absolutely continuous cdf $P(\cdot)$ and pdf $p(\cdot)$. Show that conditional distribution of $X_{(s)}$ given $X_{(1)}, \cdots\cdots, X_{(r)}$, is same as the conditional distribution of $X_{(s)}$ given $X_{(r)}$.
(This is the Markov chain property of order statistics from a continuous distribution.)

$$\rightarrow f_{X_{(r)}|X_{(s)}}(x_r|x_s) = \frac{\cancel{n!}}{(r-1)!\,(s-r-1)!\,\cancel{(n-s)!}} (F(x_r))^{r-1} (F(x_s)-F(x_r))^{s-r-1} (1-F(x_s))^{n-s} f(x_r)\, \cancel{f(x_s)}$$

$$\frac{\cancel{n!}}{(s-1)!\,\cancel{(n-s)!}} (F(x_s))^{s-1} (1-F(x_s))^{n-s} \cancel{f(x_s)}$$

$$= \frac{(s-1)!}{(r-1)!\,(s-r-1)!} \frac{(F(x_r))^{r-1} (F(x_s)-F(x_r))^{s-r-1} f(x_r)}{(F(x_s))^{s-1}}$$

$$= \frac{(s-1)!}{(r-1)!\,(s-r-1)!} \left(\frac{F(x_r)}{F(x_s)}\right)^{r-1} \left[\frac{F(x_s)-F(x_r)}{F(x_s)}\right]^{s-r-1} \frac{f(x_r)}{F(x_s)}$$

10) b) $$f_{1,2\cdots r,s}(x_1,\ldots x_r, x_s) = \frac{n!}{(s-r-1)!\,(n-s)!} f(x_1)\ldots f(x_r) \left[F(x_s)-F(x_r)\right]^{s-r-1} (1-F(x_s))^{n-r} f(x_s)$$

$$f_{1,2\ldots n} = \frac{n!}{(n-n)!} f(x_1)\cdots f(x_n)\,(1-F(x_n))^{n-n}$$

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$$\frac{X_{(r)} - \xi_p}{\sqrt{\frac{1}{4n}\frac{1}{(f(\xi_p))^2}}} \xrightarrow{L} N(0,1)$$

$$P\left[-J_{\alpha/2} \le \frac{X_{(r)} - \xi_p}{\sqrt{\frac{1}{4n}\frac{1}{(f(\xi_p))^2}}} \le J_{\alpha/2}\right] = 1-\alpha$$

$$\xi_p = X_{(r)} \pm \sqrt{\frac{1}{4n}\frac{1}{(f(\xi_p))^2}}\, J_{\alpha/2}$$

$r = np$

$[np]+1$

$$P\left[X_{(r)} \le \xi_p \le X_{(s)}\right] = \pi(r,s,n,p)$$

$r < s$

$$= \sum_{j=r}^{s-1} \binom{n}{j} p^j (1-p)^{n-j} \ge (1-\alpha)$$

$$= P\left[r \le Y \le s-1 \mid Y \sim Bin(n,p)\right]$$

$n, p = \frac{1}{2}$, $s = n-r+1 \rightarrow$ symmetric CI

$$P\left[r \le Y \le n-r \mid Y \sim Bin(n, \tfrac{1}{2})\right]$$

$$P\left[\frac{r-\frac{n}{2}}{\sqrt{\frac{n}{4}}} \le \frac{Y-\frac{n}{2}}{\sqrt{\frac{n}{4}}} \le \frac{n-r-\frac{n}{2}}{\sqrt{\frac{n}{4}}}\right] = 1-\alpha$$

finite population connection

$$P\left[\frac{r-\frac{n}{2}-\frac{1}{2}}{\sqrt{\frac{n}{4}}} \le \frac{Y-\frac{n}{2}}{\sqrt{\frac{n}{4}}} \le \frac{n-r+\frac{n}{2}+\frac{1}{2}}{\sqrt{\frac{n}{4}}}\right] = 1-\alpha$$

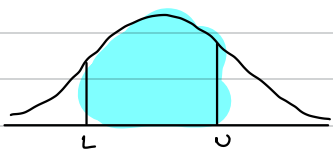
(Normal approximation for binomial)

$$\frac{r-\frac{n}{2}-\frac{1}{2}}{\sqrt{\frac{n}{4}}} = z_{\alpha/2}$$

$$\Rightarrow r = \frac{n+1}{2} - \frac{\sqrt{n}}{2} z_{\alpha/2}$$

$$n-r+1 = \frac{n+1}{2} + \frac{\sqrt{n}}{2} z_{\alpha/2}$$

Tolerance Interval / Coverage



Given $p, \gamma \in (0,1)$

$$P\left[\int_L^U f(x)\,dx \ge p\right] = \gamma \quad \left[\begin{array}{l}(U_{(r)}, U_{(s)}) \\ \text{is the } (s-r) \text{ coverage of} \\ [X_{(r)}, X_{(s)}]\end{array}\right]$$

$$= P\left[F(U) - F(L) \ge p\right] = \gamma$$

$$r < s, \quad P\left[F(X_{(s)}) - F(X_{(r)})\right] = \gamma$$

$$P\left[U_{(s)} - U_{(s)} \ge p\right] = \gamma \qquad \left(U_{(s)} - U_{(r)} \sim B_I\left(s-r, n-(s-r)+1\right)\right)$$

$$= 1 - \frac{1}{B\left(s-r, n-(s-r)+1\right)} \int_0^p u^{s-r-1} (1-u)^{n-(s-r)}\,du$$

Test for randomness :

Run Test :-

Run: 2 types of objects $\begin{cases} A \to n_1 \\ B \to n_2 \end{cases}$ $n = n_1 + n_2$

longest run: A B A ... A | B (n_1+n_2)

smallest run: A A ... A B B ... B (2)

Length of run: # identical objects in a run.

R = total no. of run's of $n = n_1+n_2$ objects.

R_1 : " " " " " n_1 items of type I.

R_2 : " " " " " n_2 " " type II.

$|R_1 - R_2| \leq 1 \Rightarrow R_1 = R_2,\ R_2-1,\ R_2+1$

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n_1 obj of type I - A, R_1 runs
n_2 " " type II - B, R_2 runs

$n_1+n_2=n$, $R = R_1+R_2$, $|R_1-R_2| \leq 1$

Test based on total no. of runs

total no. of arrangements of n_1 like objects

n_1 cells st no cells empty $= \binom{n_1-1}{n_1-1}$

$$P(R_1 = r_1,\ R_2 = r_2) = \frac{2\binom{n_1-1}{r_1-1}\binom{n_2-1}{r_2-1}}{\binom{n_1+n_2}{n_1}}$$

$r_1 = r_2$

$r_1 = r_2 - 1,\ r_2 + 1$

$$\frac{\binom{n_1-1}{r_1-1}\binom{n_2-1}{r_2-1}}{\binom{n_1+n_2}{n_1}}$$

$$P(R_1 = r_1) = \sum_{r_2} P(R_1 = r_1,\ R_2 = r_2)$$

$$\binom{n_1+n_2}{n_2} P(R_1 = r_1) = 2\binom{n_1-1}{r_1-1}\binom{n_2-1}{r_1-1} + \binom{n_1-1}{r_1-1}\binom{n_2-1}{r_1-2} + \binom{n_1-1}{r_1-1}\binom{n_2-1}{r_1}$$

$$= \binom{n_1-1}{r_1-1}\left[\binom{n_2-1}{r_1-1} + \binom{n_2-1}{r_1-1} + \binom{n_2-1}{r_1-1} + \binom{n_2-1}{r_1}\right]$$

$$= \binom{n_1-1}{r_1-1}\left[\binom{n_2}{r_1-1} + \binom{n_2}{r_1}\right]$$

$$= \binom{n_1-1}{r_1-1}\binom{n_2+1}{r_1}, \quad r_1 = 1, 2, \ldots \min\{n_1, n_2+1\}$$

$$P(R_2 = r_2) = \frac{\binom{n_2-1}{r_2-1}\binom{n_1+1}{r_2}}{\binom{n_1+n_2}{n_1}}, \quad r_2 = 1, 2 \ldots \min(n_1+1, n_2)$$

$$\binom{n_1+n_2}{n_1} P(R=r) = 2\binom{n_1-1}{\frac{r}{2}-1}\binom{n_2-1}{\frac{r}{2}-1}, \quad r_1=r_2=\frac{r}{2}, \quad r$$

$$= \binom{n_1-1}{\frac{r-1}{2}}\binom{n_2-1}{\frac{r-3}{2}} \qquad r_1=r_2-1, \quad r_2+1$$

$$+ \binom{n_1-1}{\frac{r-3}{2}}\binom{n_2-1}{\frac{r-1}{2}}$$

$r = 2(1)2\min(n_1,n_2)+c$

↳ $c=0$ if r even

$c=1$ if r odd

H_0: arrangement in random

$n=9, \quad n_1=5, \quad n_2=4, \quad \alpha=0.05$

ω: obs. $R \le c_1$ or obs. $R \geq c_2$

$\exists \; P_{H_0}(R \le c_1) + P_{H_0}(R \geq c_2) \le \alpha = 0.05$

$r = 2(1)2\min(5,4)+1$

$2(1)9$

$P_{H_0}(R \le 2) + P_{H_0}(R \geq 9) = \frac{3}{126} = 0.024 < 0.05$

$P_{H_0}(R \le 3) + P_{H_0}(R \geq 9)$

$P_{H_0}(R \le 2) + P_{H_0}(R \geq 8)$